The empirical recurrence for OEIS A299671, recovered from its tail

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Abstract

OEIS A299671 records "Empirical recurrence of order 65" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 67$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 67$. Its last coefficient is $c_{65} = -518400 \neq 0$, so no further tail satisfies anything shorter; any order-65 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299671 is "Number of nX4 0..1 arrays with every element equal to 0, 1, 2, 3, 5, 6 or 7 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$8, 88, 372, 2009, 12076, 69614, 398314, 2305246, \ \ldots$$

The entry states:

> Empirical recurrence of order 65 (see link above)

As of the "Last modified" line on the live entry (Feb 16 2018 07:23:28, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 67$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 67$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 65: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 65.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 134$ exact terms from $n = 2$ on, it returns order 65 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{65} c_i\, a(n-i),$$

c_1	9	c_{18}	1348232	c_{35}	−143783739	c_{52}	25349140
c_2	−22	c_{19}	−1968013	c_{36}	−478917464	c_{53}	102550384
c_3	70	c_{20}	5433746	c_{37}	675880675	c_{54}	−281926768
c_4	−383	c_{21}	−3819601	c_{38}	−859368774	c_{55}	74535178
c_5	597	c_{22}	1244471	c_{39}	884723499	c_{56}	141957070
c_6	−1280	c_{23}	−9496934	c_{40}	−1228978705	c_{57}	−23293630
c_7	6015	c_{24}	−20015724	c_{41}	1421056522	c_{58}	−22987012
c_8	−4714	c_{25}	5937506	c_{42}	−312267054	c_{59}	−38840108
c_9	10453	c_{26}	−47031482	c_{43}	−1466286343	c_{60}	−7072828
c_{10}	−44700	c_{27}	85699960	c_{44}	1022572726	c_{61}	3428000
c_{11}	2105	c_{28}	−46576057	c_{45}	−1099441181	c_{62}	5304216
c_{12}	−50661	c_{29}	88973591	c_{46}	−255564986	c_{63}	3331024
c_{13}	136487	c_{30}	35503319	c_{47}	151433905	c_{64}	−1180800
c_{14}	141018	c_{31}	−66210240	c_{48}	411864947	c_{65}	−518400
c_{15}	45470	c_{32}	269058865	c_{49}	−212026585		
c_{16}	215525	c_{33}	−410640029	c_{50}	49737832		
c_{17}	−1081512	c_{34}	513987836	c_{51}	224635785		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 67$, and it is the recurrence OEIS A299671 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{65} - c_1 x^{65-1} - \cdots - c_{65}$. Its constant term is $-c_{65} = 518400$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 65 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 65, so $\deg q = 65$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 65, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 65, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -518400.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-65 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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